

③ Master Theorem

$$T(n) = a T\left(\frac{n}{b}\right) + f(n)$$

$$a \geq 1, b > 1$$

Solution is

$$T(n) = n^{\log_b a} [U(n)]$$

→ $U(n)$ depends on $h(n)$

$$\rightarrow h(n) = \frac{f(n)}{n^{\log_b a}}$$

→ Relation between $h(n)$ and $U(n)$ is :

if $h(n)$	then $U(n)$
① $n^r, r > 0$	$O(n^r)$
② $n^r, r < 0$	$O(1)$
③ $(\log n)^i, i \geq 0$	$\frac{(\log_2 n)^{i+1}}{i+1}$

Problem 1

$$T(n) = 8 T\left(\frac{n}{2}\right) + n^2 \text{ --- (1)}$$

Solution

$$a=8; \quad b=2; \quad f(n)=n^2$$

$$T(n) = n^{\log_b a} [U(n)] \text{ --- (2)}$$

$$= n^{\log_2 8} [U(n)]$$

$$= n^{\log_2 2^3} [U(n)]$$

$$= n^{3 \log_2 2} [U(n)]$$

$$= n^{3 \cdot 1} [U(n)]$$

$$= n^3 [U(n)] \text{ --- (3)}$$

Where $U(n)$ depends on $h(n)$.

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{n^2}{n^3} = \frac{1}{n}$$
$$= \boxed{\frac{-1}{n}}$$

It satisfies the second condition,
because in $\frac{-1}{n}$, $\boxed{\gamma = -1}$. So
the time complexity is $O(1)$.

Apply in eqn (3), we get

$$T(n) = n^3 \cdot [U(n)]$$

$$= n^3 \cdot 1$$

$$= n^3$$

$\therefore$ So, the time complexity is $O(n^3)$

Problem 2

$$T(n) = T\left(\frac{n}{2}\right) + c \quad \text{--- (1)}$$

Solution
 $a=1, b=2, f(n)=c$

$$T(n) = n^{\log_b a} [U(n)]$$

$$= n^{\log_2 1} [U(n)] \quad \text{where } \log_2 1 = 0$$

$$= n^0 [U(n)]$$

$$= 1 \cdot [U(n)]$$

$$= [U(n)]$$

$U(n)$ depends on $f(n)$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{c}{n^{\log_2 1}} = \frac{c}{n^0} = \frac{c}{1} = \boxed{c}$$

It satisfies the third condition,

$$= \frac{(\log_2 n)^{0+1}}{0+1} \cdot c$$

$$= \frac{(\log_2 n)^1}{1} \cdot c$$

$$= \log_2 n \cdot c$$

Ignore the constant, the time complexity is $O(\log_2 n)$

Problem 3

$$T(n) = T(\sqrt{n}) + \log n \text{ if } n \geq 2$$

General form

$$T(n) = a T\left(\frac{n}{b}\right) + n^k \log^p n$$

Solution:

$$T(n) = T(\sqrt{n}) + \log n \quad \text{--- (1)}$$

Here $a=1$, $b=1$ which does not satisfy the condition $a \geq 1$, $b > 1$

So, we cannot apply master theorem directly. We should satisfy the condition $a \geq 1, b > 1$

Consider $n = 2^m$, apply in eqn (1)

$$\begin{aligned} T(2^m) &= T(2^m)^{1/2} + \log 2^m \\ &= T(2^{m/2}) + \log 2^m \end{aligned}$$

$$T(2^m) = T(2^{m/2}) + m$$

Take, $T(2^m) = S(m)$

$$S(m) = S\left(\frac{m}{2}\right) + m$$

Now $a = 1$

$b = 2$

$$f(n) = m$$

The meaning doesn't change here. Because when you form the general form

$$\begin{aligned} & n^k \log^p n \text{ where } k=1, p=0 \\ &= n^1 \log^0 n \\ &= n \cdot 1 \end{aligned}$$

$= n$ The meaning doesn't change. Just we have replaced n by m .

Consider

$$= \log 2^m$$

$$= m \log_2 2$$

$$= m \cdot 1$$

$$= m$$

Not All the conditions are satisfied, we can apply master theorem now.

$$S(m) = S\left(\frac{m}{2}\right) + m$$

$$g(m) = m^{\log_b a} \cdot [U(m)]$$

$$= m^{\log_2 1} \cdot [U(m)]$$

$$= m^0 \cdot [U(m)]$$

$$= 1 \cdot [U(m)]$$

$$= [U(m)]$$

$U(m)$ depends on $h(m)$

$$h(m) = \frac{f(m)}{n^{\log_b a}} = \frac{m}{n^{\log_2 1}} = \frac{m}{n^0}$$

$$= \frac{m}{1} = \boxed{m}$$

Now, find the value of m ,

$$n = 2^m$$

Apply $\log$ on both sides

$$\log n = \log 2^m$$

$$\log n = m \log_2 2$$

$$\log n = m \cdot 1$$

$$\boxed{m = \log_2 n}$$

It satisfies the third condition

$$= \frac{(\log_2 m)}{i+1}$$

$$= \frac{(\log_2^n)}{0+1}$$

$$= \frac{(\log_2^n)}{1}$$

$$= \boxed{\log_2^n}$$

So, the time complexity is

$$\boxed{O(\log_2^n)}$$